

Flurry Analytics

Android SDK Instructions

SDK version 6.5.0
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Welcome to Flurry Analytics!

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1. Introduction

The Flurry Android Analytics Agent allows you to track the usage and behavior of your Android application on users' phones for viewing in the Flurry Analytics system. It is designed to be as easy as possible with a basic setup complete in few minutes.

Flurry Analytics uses the Android Advertising ID provided by Google Play Services and will check for and respect the user's ad tracking preference. For more information, please visit https://developer.android.com/google/play-services/id.html

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2. Integration

Flurry Analytics requires minimum Android API level 10. To integrate Flurry Analytics into your Android application:

Step 1. Add the FlurryAnalytics_6.5.0.jar file to your classpath.

Using Maven dependencies:

```
buildscript {
        repositories {
                jcenter()
        }
}

dependencies {
        compile 'com.flurry.android:analytics:6.5.0'
```

}

Using Jar files:

Using Android Studio:

1. Add FlurryAnalytics-6.5.0.jar to your project's libs folder.
2. Navigate to File > Project Structure > Module > Dependencies. Click the '+' button in the bottom of the 'Project Structure' popup to add dependencies. Select 'File dependency' and add libs/FlurryAnalytics-6.5.0.jar.
3. Add Google Play Services library. Please follow instructions at http://developer.android.com/google/play-services/setup.html#Setup
4. Add v4 support library (or greater). Please follow instructions at http://developer.android.com/tools/support-library/features.html#v4

Using Eclipse:

1. Add FlurryAnalytics-6.5.0.jar to your project's libs folder. Right click on each JAR file and select Build Path > Add to Build Path.
2. Add Google Play Service library. Please follow instructions at http://developer.android.com/google/play-services/setup.html#Setup
3. Add v4/v7 Support Library. Please follow instructions at https://developer.android.com/tools/support-library/setup.html#add-library.

Step 2. Configure AndroidManifest.xml:

Required Permission:

`android.permission.INTERNET`

Required to send analytics data back to the flurry servers

Optional Permission (Highly Recommended):

`android.permission.ACCESS_NETWORK_STATE`

If your application has network state permissions, transmission of analytics data can be optimized.

Optional Permission:

`android.permission.ACCESS_COARSE_LOCATION` or
`android.permission.ACCESS_FINE_LOCATION`

If your application has location `android.permission.ACCESS_FINE_LOCATION` permission, analytics will use `PASSIVE_PROVIDER` to get the location information. Otherwise it will use the `NETWORK_PROVIDER` to fetch the last known location.

Without these permissions, only country level location information will be available. You may call `FlurryAgent.setLocation()` to set the location manually if your app uses the GPS. To disable detailed location reporting even when your app has permission, call `FlurryAgent.setReportLocation(false)` before calling `FlurryAgent.onStartSession()`

Specify a versionName attribute in the manifest to have data reported under that version name.

Application:

If you are shipping an app, extend the Application class if you are not already doing so:

```
<application
  android:name=".MyApplication"
  …
</application>
```

Step 3. Add calls to init, onStartSession and onEndSession

1. If you are shipping an app, use the FlurryAgent Builder to init the SDK, passing it a reference to your Application Context and your project's API key:

   ```
   public class MyApplication extends Application {
     @Override
     public void onCreate() {
       super.onCreate();

        // configure and init Flurry
       new FlurryAgent.Builder()
               .withListener(flurryAgentListener)
               .withLogEnabled(true)
               .withLogLevel(Log.INFO)
               .withContinueSessionMillis(5000L)
               .withCaptureUncaughtExceptions(false)
               .withPulseEnabled(true)
               .build(context, "apiKey");
     }
   }
   ```

   It is safe to call the Builder more than once, provided that you use the same API key throughout the application. You may use any type of Context you wish. The following methods are now deprecated: setFlurryAgentListener(), setLogEnabled(), setLogLevel(), setContinueSessionMillis(), setCaptureUncaughtExceptions(), setPulseEnabled() and init().

2. If you are writing an app and the minimum target is Ice Cream Sandwich or above (minSdkVersion is set to API level 14 or greater), session handling is completely automatic and you may skip steps 3 and 4. If you are instrumenting another type of Context, such as a Service, or your minimum target is Gingerbread, proceed with steps 3 or 4.

3. Insert a call to `FlurryAgent.onStartSession(Context)` passing it a reference to a Context object (such as an Activity or Service). If you are targeting Gingerbread, we recommend using the onStart method of each Activity in your application, and passing the Activity itself as the Context object. For services (or other Contexts), use the Service or the relevant Context as the Context object. Do not pass in the global Application context.

4. Insert a call to `FlurryAgent.onEndSession(Context)`when a session is complete. If you are targeting Gingerbread, we recommend using the onStop method of each Activity in your application.

For services (or other Contexts), ensure that onStop is called in each instrumented Service. Make sure to match up a call to onEndSession for each call of onStartSession, passing in the same Context object that was used to call onStartSession. Do not pass in the global Application context.

So long as there is any Context that has called onStartSession but not onEndSession, the session will be continued. Also, if a new Context calls onStartSession within 10 seconds of the last Context calling onEndSession, then the session will be resumed, instead of a new session being created. Session length, usage frequency, events and errors will continue to be tracked as part of the same session. This ensures that as a user transitions from one Activity to another in your application that they will not have a separate session tracked for each Activity, but will have a single session that spans many activities. If you want to track Activity usage, we recommend using logEvent, described below.

If you wish to change the window during which a session can be resumed, call `FlurryAgent.setContinueSessionMillis(long milliseconds)` before the call to FlurryAgent.init(). **This API is now deprecated, to use the new API, please refer to Section 3.1.**

You're done! That's all you need to do to begin receiving basic metric data.

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3. Optional Features

Report Additional Data
You can use the following methods (during a session only) to report additional data:

```
FlurryAgent.logEvent(String eventId)
FlurryAgent.logEvent(String eventId, boolean timed)
FlurryAgent.logEvent(String eventId, Map<String, String> parameters)
FlurryAgent.logEvent(String eventId, Map<String, String> parameters, boolean
timed)
```

Use FlurryAgent.logEvent to track user events that happen during a session. You can track how many times each event occurs, what order events happen in, how long events are, as well as what the most common parameters are for each event. This can be useful for measuring how often users take various actions, or what sequences of actions they usually perform. Each project supports a maximum of 300 event names, and each event id, parameter key, and parameter value must be no more than 255 characters in length. Each event can have no more than 10 parameters. The parameter argument is optional, and may be null. Each session can log up to 200 events and up to 100 unique event names.

```
FlurryAgent.endTimedEvent(String eventId)
FlurryAgent.endTimedEvent(String eventId, Map<String, String> parameters)
```

Timed event can be logged using FlurryAgent.logEvent. Use endTimedEvent to end the timed event.

```
FlurryAgent.onError(String errorId, String message, Throwable exception)
```

Use onError to report errors that your application catches. Flurry will report the first 10 errors to occur in each session.

```
FlurryAgent.setCaptureUncaughtExceptions(false)
```

Used to allow/disallow Flurry SDK to report uncaught exceptions. The feature is enabled by default and if you would like to disable this behavior, this must be called before calling `init`. **This API is now deprecated, to use the new API, please refer to Section 3.1.**

```
FlurryAgent.onPageView()
```

Use onPageView to report page view count. You should call this method whenever a new page is shown to the user to increment the total count. Page view is tracked separately from events.

`FlurryAgent.setLogEvents(boolean logEvents)`
Use setLogEvents to enable/disable the event logging. This should be called before `init`.

`FlurryAgent.setReportLocation(boolean)`
Use setReportLocation to enable/disable location tracking. Flurry uses cached value (to avoid excessive battery usage) when location reporting is enabled. You may call `FlurryAgent.setLocation()` to set the location manually if your app uses the GPS.

Pulse Monitoring
`FlurryAgent.setPulseEnabled(boolean enablePulse)`

Enabling Flurry Pulse shares your app data with Pulse integrated partners. Starting version 5.5.0, Flurry provides means to turn on sending selected session events to integrated partners. You need to turn on Pulse using the API call before calling `FlurryAgent.init()`. Use `setPulseEnabled` to enable/disable Flurry Pulse. **This API is now deprecated, to use the new API, please refer to Section 3.1.**

Flurry SDK automatically queries the config end points selected by you and send url's appropriately. SDK does not send any data to partners unless you have set data delivery rules on Pulse UI.
Note:

Pulse integrated partners may require install referrer information. If you are planning to share data with such partners, you must pass the install referrer information to the Flurry SDK by one of two following options:

1. If your app is not using a custom install referrer receiver, declare ours in your manifest
   <receiver
   android:name="com.flurry.android.FlurryInstallReceiver"
   android:exported="true" >
   <intent-filter>
   <action android:name="com.android.vending.INSTALL_REFERRER" />
   </intent-filter>
   </receiver>

2. If your app is using a custom install referrer receiver, then manually call our receiver in your onReceive method.
   FlurryInstallReceiver flurryInstallReceiver = new InstallReceiver();
   flurryInstallReceiver.onReceive(context, intent);

Session Information

`FlurryAgent.setFlurryAgentListener(FlurryAgentListener)`
Sets the listener to receive callback methods when session is started or finished. It is recommended to call this method before `FlurryAgent.init()`
Callback method `onSessionStarted` is received when Flurry session starts. **This API is now deprecated, to use the new API, please refer to Section 3.1.**

`FlurryAgent.isSessionActive()`
Returns true if Flurry session is active and `false` otherwise.

`FlurryAgent.getSessionId()`
Gets the active Flurry session id.

Tracking Demographics

`FlurryAgent.setUserID(String);`
Use this to log the user's assigned ID or username in your system. This should be called before `init`, if possible.

`FlurryAgent.setAge(int);`
Use this to log the user's age. Valid inputs are between 1 and 109. This should be called before `init`, if possible.

`FlurryAgent.setGender(byte);`
Use this to log the user's gender. Valid inputs are `Constants.MALE`or `Constants.FEMALE` This should be called before `init`, if possible.

`FlurryAgent.setLocationCriteria(Criteria locationCriteria)`
This method is deprecated in SDK version 5.3.0. Flurry SDK now uses PASSIVE_PROVIDER to fetch the location information if app has ACCESS_FINE_LOCATION permission. Otherwise it will use the NETWORK_PROVIDER to fetch the last known location.

Additional Features
`FlurryAgent.getReleaseVersion()`
Returns a string containing current Flurry SDK release version

`FlurryAgent.setVersionName(String versionName)`
Sets the version name of the app. This name will appear in the http://dev.flurry.com as a filtering option by version. This should be called before `init`.

`FlurryAgent.getAgentVersion()`
Gets the version of the Flurry SDK.

`FlurryAgent.setLocation(float latitude, float longitude)`
Call this method to set the current location (used with geographical targeting). This should be called before `init`, if possible; otherwise call it as soon as the location is available.

`FlurryAgent.setLogLevel(int logLevel)`
Sets the log level of the internal Flurry SDK logging. Valid inputs are Log.VERBOSE, Log.WARN etc. Default log level is Log.WARN. This should be called before `init`. **This API is now deprecated, to use the new API, please refer to Section 3.1.**

`FlurryAgent.setLogEnabled(boolean isEnabled)`
Use `setLogEnabled` to enable/disable internal Flurry SDK logging. This should be called before `init`.
This API is now deprecated, to use the new API, please refer to Section 3.1.

`FlurryAgent.addOrigin(String originName, String originVersion)`
`FlurryAgent.addOrigin(String originName, String originVersion, Map<String, String> originParameters)`
Use addOrigin to add the origin attribution. The event is identified by the originName, originVersion and originParameters. OriginParameters can be passed in as a Map<String,String> where the key is the parameter name, and the value is the value. This should be called before `init`.

ProGuard
If you plan to run ProGuard on your APK before releasing your app, you will need to add the following to your “proguard.cfg” file:

```
-keep class com.flurry.** { *; }
-dontwarn com.flurry.**
-keepattributes *Annotation*,EnclosingMethod
-keepclasseswithmembers class * {
public <init>(android.content.Context, android.util.AttributeSet, int);
}

# Google Play Services library
-keep class * extends java.util.ListResourceBundle {
   protected Object[][] getContents();
}

-keep public class
com.google.android.gms.common.internal.safeparcel.SafeParcelable {
   public static final *** NULL;
}

-keepnames @com.google.android.gms.common.annotation.KeepName class *
-keepclassmembernames class * {
   @com.google.android.gms.common.annotation.KeepName *;
}

-keepnames class * implements android.os.Parcelable {
   public static final ** CREATOR;
}
```

Please let us know if you have any questions. If you need any help, just email androidsupport@flurry.com!

Cheers,
The Flurry Team
http://www.flurry.com
androidsupport@flurry.com